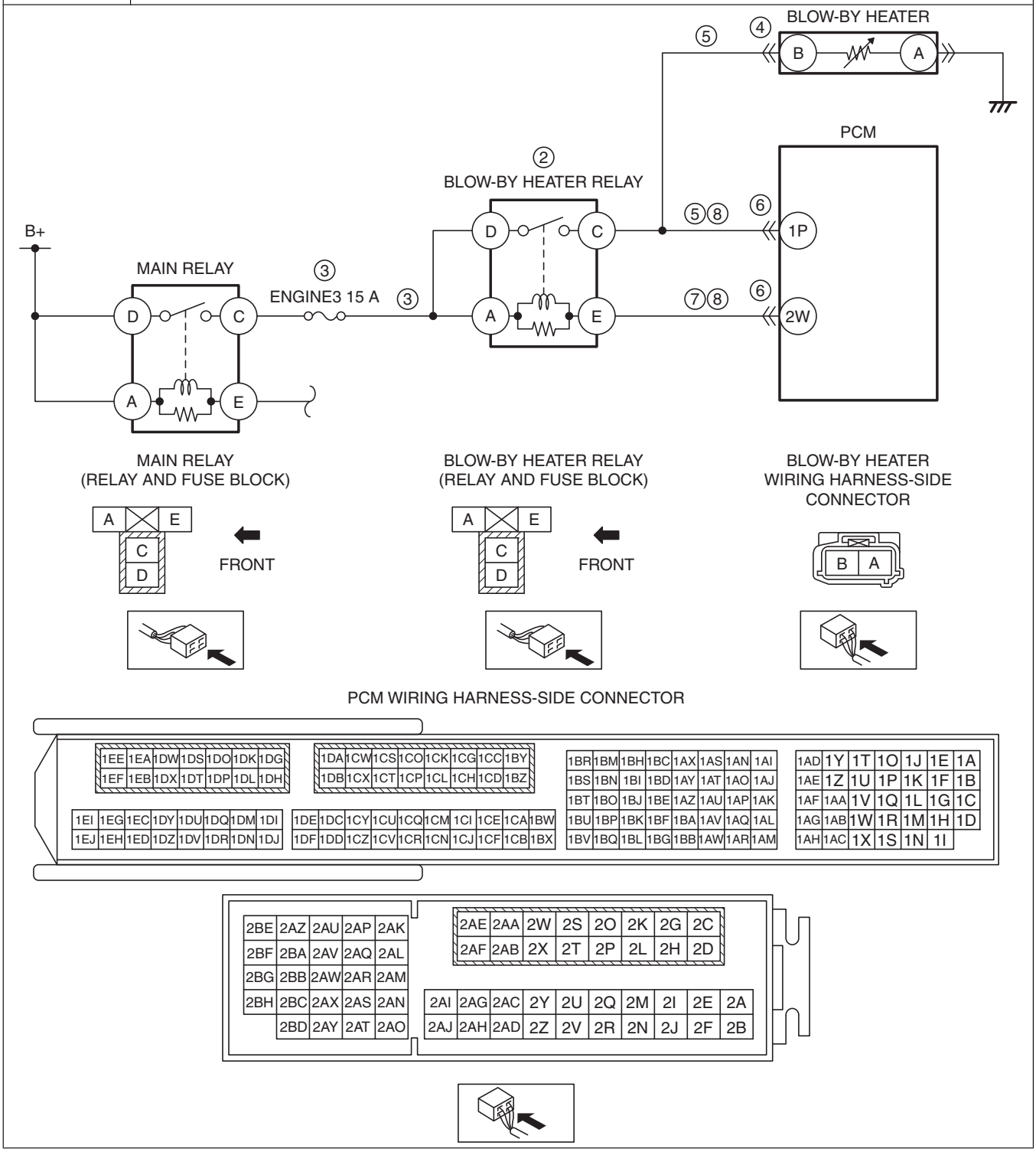
DTC P053B:00 [SKYACTIV-D 1.5]

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DTC P053B:00	Blow-by heater relay control circuit low input
DETECTION CONDITION	• With the following conditions met, the blow-by heater power supply voltage is continuously lower than the specification for **5 s**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Blow-by heater relay controlled on **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Blow-by heater relay malfunction • Short to ground or open circuit in blow-by heater relay power supply circuit — Short to ground in wiring harness between ENGINE3 15 A fuse and blow-by heater relay terminal D — Short to ground in wiring harness between ENGINE3 15 A fuse and blow-by heater relay terminal A — ENGINE3 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and blow-by heater relay terminal D — Open circuit in wiring harness between main relay terminal C and blow-by heater relay terminal A • Blow-by heater connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Blow-by heater relay terminal C—Blow-by heater terminal B — Blow-by heater relay terminal C—PCM terminal 1P • PCM connector or terminals malfunction • Short to power supply in wiring harness between blow-by heater relay terminal E and PCM terminal 2W • Open circuit in wiring harness between the following terminals: — Blow-by heater relay terminal C—PCM terminal 1P — Blow-by heater relay terminal E—PCM terminal 2W • Blow-by heater malfunction • PCM malfunction

DTC P053B: 00	Blow-by heater relay control circuit low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**INSPECT BLOW-BY HEATER RELAY** • Switch the ignition off. • Remove the blow-by heater relay. (See RELAY LOCATION.) • Inspect the blow-by heater relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the blow-by heater relay, then go to Step 10.
		No	Go to the next step.
3	**INSPECT BLOW-BY HEATER RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Blow-by heater relay is removed. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — Blow-by heater relay terminal D — Blow-by heater relay terminal A • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE3 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • ENGINE3 15 A fuse—Blow-by heater relay terminal D • ENGINE3 15 A fuse—Blow-by heater relay terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Main relay terminal C—Blow-by heater relay terminal D • Main relay terminal C—Blow-by heater relay terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
4	**INSPECT BLOW-BY HEATER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the blow-by heater connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT BLOW-BY HEATER MONITOR CIRCUIT FOR SHORT TO GROUND** • Blow-by heater relay is removed. • Verify that the blow-by heater connector is disconnected. • Inspect for continuity between blow-by heater relay terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Disconnect the PCM connector and inspect the wiring harness for short to ground. • If the short to ground circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Blow-by heater relay terminal C—Blow-by heater terminal B • Blow-by heater relay terminal C—PCM terminal 1P **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • If the short to ground circuit could not be detected in the wiring harness: — Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 10.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
7	**INSPECT BLOW-BY HEATER RELAY CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Blow-by heater relay is removed. • Verify that the blow-by heater and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the blow-by heater relay terminal E (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between blow-by heater relay terminal E and PCM terminal 2W. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
8	**INSPECT BLOW-BY HEATER RELAY CIRCUIT FOR OPEN CIRCUIT** • Blow-by heater relay is removed. • Verify that the blow-by heater and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Blow-by heater relay terminal C—PCM terminal 1P — Blow-by heater relay terminal E—PCM terminal 2W • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Blow-by heater relay terminal C—PCM terminal 1P • Blow-by heater relay terminal E—PCM terminal 2W **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT BLOW-BY HEATER** • Inspect the blow-by heater. (See BLOW-BY HEATER INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the blow-by heater, then go to the next step. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.